

SkillBridge — SIH 2026

TECHNICAL APPROACH & END-TO-END IMPLEMENTATION GUIDE

Academia-Industry Collaboration Portal • Closed-Loop Skill Matching Prototype

1. Executive Summary & Paradigm Shift

SkillBridge resolves the structural crisis in Indian graduate employability: traditional campus placement relies on unverified resume keyword filtering, encouraging candidate keyword stuffing, blinding universities to syllabus obsolescence, and forcing corporate talent teams to manually filter hundreds of unqualified candidates.

SkillBridge shifts higher education recruitment to a verified competency model: students prove capability through benchmarked assessments; an AI recommendation engine maps verified skill profiles to weighted corporate roles; borderline candidate scores (70–84.9%) are automatically escalated to faculty mentors for human review; and aggregated deficit data flows back into university curricula.

Core Innovation: The Closed-Loop Feedback Architecture

Unlike conventional ATS platforms that terminate after application submission, SkillBridge closes the institutional loop: corporate hiring requirements directly influence semester curricula, and student assessment telemetry validates real classroom competencies.

2. High-Level Technical Architecture

Layer	Technology Selection	Architectural Role & Justification
Frontend Framework	Next.js 14 (App Router & React 18)	Server Components for rapid SSR, zero layout shifts, client reactivity
Database Layer	PostgreSQL on Neon Cloud	Serverless cloud Postgres isolated in custom schema "skillbridge"
Data Access & Schema	Prisma ORM (25 Consolidated Models)	Strict type-safe schema, embedded scoring JSON, audit logs
Authentication Engine	NextAuth.js (JWT Strategy & bcrypt)	Zero-roundtrip JWT session guards, 1-click instant demo access
Styling & Visual Design	Tailwind CSS (Zero-Blue Palette)	Deep Forest Green (#1B4332), Warm Gold (#D97706), Warm Parchment
DevOps & Deployment	Vercel + GitHub Repository	Production deployment pipeline at https://github.com/Codewithkrishna987/skillbridge

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SKILL PORTAL AI IMPLEMENTATION FLOW (5 CORE PHASES)

Detailed breakdown of the continuous intelligence pipeline

3. The 5-Stage AI Intelligence Pipeline

The SkillBridge AI implementation pipeline operates continuously across five distinct phases, linking raw assessment telemetry to recruiter hiring decisions and university curriculum enhancements.

Phase 01: Skill Data Indexing

Technical Function: Taxonomy Structuring & Dynamic Vector Embeddings

Student assessment responses and industry skill requirements are structured and embedded into the PostgreSQL database. Automated synchronization jobs monitor corporate job posts to continuously refresh the standardized skill taxonomy, mapping hierarchical relationships between programming languages, frameworks, and tools.

- Impact Metric: Taxonomy Coverage: 120+ Industry Benchmarks across 8 Technology Domains

Phase 02: Assessment Processing

Technical Function: Real-Time Semantic Evaluation & Rubric Scoring

Each student's questionnaire and technical aptitude responses are semantically evaluated in real time against the industry skill taxonomy. The engine calculates verified competency percentiles (0–100%) across individual skills, identifying relative candidate strengths and quantifiable skill gap deficits.

- Impact Metric: Telemetry Precision: Sub-second response evaluation with strict rubric weighting

Phase 03: Skill Profile Generation

Technical Function: Hybrid Recommendation Engine & Gap Alerting

The recommendation engine synthesizes the student's verified competency profile with live industry opportunity data. It generates personalized learning roadmaps, course recommendations to bridge identified deficits, and ranks matching internships with transparent match percentage scores.

- Impact Metric: Explainable Fit: Full transparency on exact skill matches vs deficit gaps

Phase 04: Matching & Escalation

Technical Function: Human-in-the-Loop Academic Escalation Safeguard

High-confidence matches (>85%) are presented to students and corporate recruiters instantly. Ambiguous or borderline matches (70%–84.9%) are automatically flagged and routed to the faculty mentor or academic advisor for manual domain review and endorsement before recruiter dispatch.

- Impact Metric: Safety Threshold: Borderline candidate protection against premature algorithm rejection

Phase 05: Continuous Feedback Loop

Technical Function: Post-Placement Outcomes & Curriculum Adaptation Signal

Recruiter feedback, interview outcome data, and final placement records are fed back into the model. Aggregated industry deficits inform university academic councils and departments, generating concrete signals on which technologies must be introduced into upcoming semester syllabi.

- Impact Metric: Closed Loop: Direct pipeline from corporate hiring demand to university classroom

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END-TO-END SKILL MAPPING & PLACEMENT FLOW

9-Step Chronological Journey from Registration to Institutional Analytics

4. The 9-Step Operational Lifecycle

SkillBridge guides every participant through an end-to-end traceable journey, ensuring zero unverified claims and full academic institutional participation.

01	Student Registration Stakeholder: Student	Candidate logs into unified portal via NextAuth credentials or institutional SSO.
02	Skill Assessment Stakeholder: Student & Engine	Student takes standardized objective test catalogs (React, TypeScript, Docker, etc.).
03	Profile Generation Stakeholder: AI Pipeline	AI synthesizes assessment telemetry into a verified digital skill profile with confidence ratings.
04	Smart Matching Stakeholder: AI Pipeline	Opportunities posted by corporate partners are mapped to student profiles with match percentages.
05	Student Applies Stakeholder: Student	Student reviews transparent match scores and applies directly with verified competencies.
06	Application Tracking Stakeholder: Recruiter & Student	Portal tracks status across: Applied -> Mentor Endorsed -> Recruiter Shortlisted -> Interviewing.
07	Feedback & Portfolio Stakeholder: Mentor & Student	Faculty mentor endorsement notes and verified credentials auto-populate the digital portfolio.
08	Academician Loop Stakeholder: Academician	Professors analyze mentee performance and receive alerts on emerging industry skill deficits.
09	Institutional Analytics Stakeholder: University Dean	Aggregated university dashboards show placement rates, cohort readiness, and curriculum drift.

5. Stakeholder Portal Inventory (17 Operational Routes)

- **Student Portal:** /student, /assessments, /skill-gaps, /opportunities, /applications, /portfolio
- **Academician Portal:** /academician, /mentees, /reviews (Escalated Queue), /verifications, /collaborations
- **Industry Recruiter:** /industry, /post, /applications (Ranked Pipeline), /skill-demand
- **Admin Console:** /admin, /directory, /skills, /assessments, /audits

6. 5-Minute Judge Presentation Script

1. The Paradigm Pitch (Landing Page)

Route: <http://localhost:3000>

Open the homepage. Walk judges through the 5-Stage AI Implementation Flow and 9-Step End-to-End Placement Flow matching the national presentation slide. Explain how resume keyword filtering is replaced by verified competency scoring.

2. 1-Click Instant Demo Login (Zero Typing)

Route: <http://localhost:3000/login>

Click "Student: Aarav Mehta". Authenticate instantly into the Student Workspace. Show verified competencies (React 90%, TypeScript 85%) and the 55% Docker deficit alert in Skill Gap Analysis.

3. Smart Matching & Phase 4 Escalation Trigger

Route: <http://localhost:3000/student/opportunities>

Show the Nexura Cloud internship with an 84.5% Match Score. Click "Apply with Verified Score". Show the modal explaining how borderline scores (70–84.9%) automatically escalate to faculty mentors.

4. Faculty Mentor Endorsement in Real Time

Route: <http://localhost:3000/academician/reviews>

Switch to Academician (Dr. Aris Thorne). Open the Escalated Match Review Queue. Click "Endorse Candidate". The application is stamped with the official faculty endorsement seal and forwarded to the recruiter.

5. Recruiter Pipeline & Institutional Analytics

Route: <http://localhost:3000/industry/applications>

Switch to Recruiter. View the candidate pipeline ordered by verified competency. Show the "Faculty Endorsed" badge and click "Advance to Interview". Then view the Academician dashboard to show the curriculum gap advisory.

7. Next Steps & Resources Required from User

Everything in the SkillBridge prototype is currently 100% operational, fully seeded, and running locally on port 3000 with live Neon PostgreSQL connectivity.

- **Immediate Action** Run the 5-minute judge demo script above during your SIH evaluation session.
- **Print / Export** Both this PDF and the interactive web view at <http://localhost:3000/prototype-doc> are ready for printing.
- **GitHub Repository** Your code is committed and pushed to <https://github.com/Codewithkrishna987/skillbridge>.
- **Resources Required** None for local demo! All database connections, auth users, and mock heuristics are self-contained.
- **Optional Cloud Deployment** If you wish to deploy live to Vercel, provide a Vercel token or link your GitHub repo directly in vercel.com.